G-Speak

G-Speak, was developed by John Underkoffler. Underkoffler served as a technical advisor for Steven Spielberg's film "Minority Report" where the system was used by Tom Cruise. Its a system which allows users to navigate and interact with data in an unprecedented, visually rich, natural and responsive manner. It would interpret a user's motion to move through datasets without him or her needing to use a computer mouse or any other physical object to do so. It uses a mounted camera to track the user's gloves as they move in space. It has sub millimeter precision and can handle quick motions and multiple users. It also has an I/O Bulb is based on a traditional light bulb but is able to not only project light, but also collect live video of the objects and surfaces it projects light onto.

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wrist piece in a particular pattern to indicate that he wanted to buy their product. Transactions were now simplified down to a swipe. Money was now simply a number associated with a person or institution. As he put on his sun-glasses, the HUD displayed an plethora of information, including the weather, location aware content such as advertisements of beverages as he passed by a pub, online activities of his friends living in the vicinity and reminders of his to-do list at that time.

All of this was powered by his personal computer a chip that was responsible for the processing power of his HUD glasses and his wrist-piece where it was embedded. It was the age where devices were increasingly personal and collaborative at the same time.

The cities were designed using actual miniature models using technologies like Sandscape. With the advent of Sandscape, people was able to use actual sand to model structures and have a computer churn out topographical information, height, drainage, and contours as coloured projections on it.

Nothing like Home

Back at Achal's home, his mother, was trying on dresses in front of the mirror. It displayed information of the cloth and recommended her an attire based on her catalogue, previous preferences and present mood (using technologies such as Q™ Sensor, a wireless, wearable biosensor that measures a key physical component of emotions, developed way back in 2011).

His dad while returning home in the evening, set the temperature of his room remotely courtesy his mobile device and the city wide network grid. As he reached the front door he identified himself to the security system. All he had to do was to wear a headpiece and concentrate on opening the door. Brain Machine Interfacing(BMI) had evolved into an exact science. Simple things like turning on or off the lights, opening or closing the doors and windows, turning devices on or off, moving a wheelchair, assigning commands to a robot etc. was just a matter of thinking that command in your mind. With BMI the way we interface with technology is- no interface at all; it is quite literally, Mind over Matter.

The nanotech enabled house had furniture which remembered every member's ergonomic profiles to shift its shape accordingly. It folded and hid when not in use. The walls changed colour. Every device in the house could communicate with each other. Every house was self sufficient with their own energy producing box. Every object was electronically enabled, that is the dining table was aware of what was placed on it etc.

In the future, broadly, all technology will eventually fall into two categories: network, and interface. One will connect device with device while the other will connect a device with a human. The progress in how humans interface with technology will harbour its deeper integration into our daily lives; and it is this very integration that will lead to technology fulfilling its goal of improving the quality of the human experience for the average joe.

Brain Machine Interfacing

Products like The Intendix from Guger Technologies (g*tec) and EPOC headset have already been able to achieve a non-invasive BMI, that is, there is the sensors dont need to be inside the skull. The EPOC is a streamlined EEG device that translates your thoughts and facial expressions into computer inputs. In BCI systems, electrodes sense the waves of the brain and software interprets it, finally executing that action.